

Literature Synthesis: Sociotechnical AI Audits

Working Group on Sociotechnical AI Auditing

79 papers reviewed | May 2026 | Draft for working group discussion

1. Overview and Category Distribution

This synthesis covers 78 papers collected for the working group on sociotechnical AI audits. Each paper has been assigned to one of seven thematic buckets. The distribution is as follows:

Category	Count	Share
Sociotechnical/impact-focused audits	27	35%
Organizational/process audits	19	24%
Other (ethics frameworks, conceptual)	13	17%
Technical/model auditing	10	13%
Regulatory/compliance audits	5	6%
Industry frameworks (Big 4, NIST, etc.)	4	5%

2. Within-Category Key Themes

2.1 Technical/Model Auditing (10 papers)

What do these papers audit?

Papers in this category focus on the properties and outputs of AI models themselves. The primary audit targets are fairness and bias in model predictions (Silicon Ceiling, Empirical Likelihood-Based Fairness Auditing, Auditing the AI Auditors), content moderation behavior (Auditing GPT's Content Moderation), dialect-based quality-of-service disparities (Framework for Auditing Chatbots), privacy guarantees in differentially private ML (Privacy Auditing in Differential Private ML), and the reliability of evaluation benchmarks (Can we trust AI benchmarks?). The LLMAuditor paper targets LLM outputs more broadly using human-in-the-loop probe generation.

How do they define 'audit'?

Most papers in this bucket treat auditing as systematic, empirical testing of model behavior against some normative standard—typically a fairness metric or performance benchmark. The framing is often quasi-experimental: generate inputs, observe outputs, measure disparities. Several papers explicitly draw on the tradition of social science audit studies (Silicon Ceiling models its approach on resume audit methodology). The AI benchmarks meta-review treats evaluation itself as an object of scrutiny, essentially auditing the audit tools.

Methods

The dominant methods are quantitative: statistical fairness metrics (demographic parity, equalized odds, calibration), hypothesis testing frameworks (Empirical Likelihood-Based), scaling laws for resource-constrained settings (Audits Under Resource Constraints), and systematic prompt-based probing (LLMAuditor, Auditing GPT Content Moderation). The benchmarks review takes a meta-analytical approach, cataloguing design flaws across ~110 evaluation studies.

What do they miss?

The consistent gap is context. Technical audits tend to evaluate models in isolation, abstracting away the deployment environment, organizational incentives, and affected communities. Fairness metrics are treated as given rather than contested. There is minimal attention to who selects the metrics, which populations are represented in test data, or how audit findings translate into organizational action. The LLMAuditor and Chatbot Dialect papers begin to address this by incorporating human judgment, but even these remain primarily model-focused.

Key assumptions

These papers generally assume that (a) the relevant harms can be quantified through model outputs, (b) fairness can be operationalized through statistical measures, (c) test-time evaluation is a reasonable proxy for real-world behavior, and (d) auditors have sufficient access to probe the model meaningfully. The benchmarks paper directly challenges assumption (c),

showing that one-time testing logic fails to account for how models interact with humans and other systems.

2.2 Organizational/Process Audits (19 papers)

What do these papers audit?

Papers here target organizations, governance structures, and the auditing process itself rather than specific models. Audit objects include: internal governance and documentation practices (Closing the AI Accountability Gap, Operationalising AI governance), the landscape of available audit tools and methods (Towards AI Accountability Infrastructure, From ethical AI frameworks to tools, Applying the ethics of AI), access and evidence frameworks for auditors (From Transparency to Accountability, Black-Box Access is Insufficient), reporting and contextualization standards (Audit Cards, Model Openness Framework), and the emerging AI auditing ecosystem as a whole (The emergence of AI ethics auditing, AI auditing: The Broken Bus, Auditing of AI: Legal Ethical Technical).

How do they define 'audit'?

Definitions here are more varied and reflective. Several papers explicitly distinguish between technology-oriented audits (evaluating model properties) and process-oriented audits (evaluating governance structures). The Broken Bus paper offers perhaps the most critical definition, finding that only a subset of audit practices translate to actual accountability outcomes. The AstraZeneca case study reveals that 'audit' in practice involves navigating classical organizational challenges: harmonizing standards across decentralized units, scoping the audit, driving change management, and measuring outcomes.

Methods

Methods are diverse: systematic literature reviews (Ethics-based AI auditing SLR, From ethical AI frameworks to tools), qualitative interviews (The emergence of AI ethics auditing), longitudinal case studies (Operationalising AI governance), framework proposals (Closing the AI Accountability Gap, Auditing LLMs three-layered approach), and gap analyses (Towards AI Accountability Infrastructure). Several papers propose specific audit artifacts: Audit Cards for structured reporting, the Model Openness Framework for transparency classification, and the Shneiderman paper for multi-level governance recommendations.

What do they miss?

While these papers are more attentive to organizational realities than the technical category, they tend to underspecify the power dynamics that shape audit outcomes. Who commissions the audit, who has veto power over findings, and what enforcement mechanisms exist are rarely addressed in depth. The Broken Bus paper is a notable exception. Additionally, many framework proposals remain theoretical and have not been empirically validated. The gap between proposing an audit process and demonstrating that it works in practice is substantial.

2.3 Regulatory/Compliance Audits (5 papers)

What do these papers audit?

This category addresses the intersection of AI auditing with legal and regulatory frameworks. Papers examine the EU's regulatory landscape (Addressing the regulatory gap, How to investigate algorithmic-driven risks under the DSA), government AI regulation challenges (Regulating Government AI), EU ethics guidelines as benchmarks (Establishing the Rules for Building Trustworthy AI), and transparency/accountability in the context of legal frameworks (Transparency and accountability in AI systems).

How do they define 'audit'?

Audit here is framed primarily as compliance verification and regulatory oversight. The DSA paper develops a taxonomy of study designs for investigating algorithmic risks within regulatory requirements. The regulatory gap paper defines an AI audit ecosystem that incorporates both compliance and oversight functions, arguing that civil society must be included. These papers treat auditing as a legal-institutional mechanism rather than a purely technical or organizational one.

What do they miss?

The regulatory papers tend to be Europe-centric, reflecting the EU's position as the leading AI regulatory jurisdiction. There is limited coverage of regulatory approaches in the US, China, or the Global South. The papers also tend to assume that regulation, once enacted, will be effectively implemented and enforced, without deeply examining the capacity constraints, political dynamics, and industry lobbying that shape regulatory outcomes in practice.

2.4 Sociotechnical/Impact-Focused Audits (27 papers)

What do these papers audit?

This is the largest and most heterogeneous category. Papers range from conceptual frameworks for sociotechnical AI governance (A sociotechnical system perspective on AI, Sociotechnical AI Governance: Challenges and Opportunities for HCI) to empirical audits of specific systems (A Sociotechnical Audit: Assessing Police Use of Facial Recognition, Sociotechnical Audits: Broadening the Algorithm Auditing Lens). Many papers propose frameworks that integrate technical and social dimensions: the three-layered sociotechnical safety evaluation of generative AI (Weidinger et al.), the multi-component bias framework (Fairness and bias in AI: a sociotechnical perspective), and the decolonial approach (Decolonial AI). Several address specific domains: cybersecurity (two papers), healthcare (Shaping the Future of Healthcare), and AI decommissioning (The AI Sunset Protocol).

How do they define 'audit'?

The defining feature of this category is an expansive notion of auditing that encompasses not just technical performance but the full sociotechnical system: stakeholder impacts, power dynamics, contextual factors, organizational structures, and societal consequences. Many papers frame auditing as an ongoing process of sense-making and accountability rather than a one-time compliance check. The Weidinger et al. paper articulates this most explicitly with its three layers (capability evaluation, human interaction, systemic impacts), arguing that context determines whether a capability causes harm.

Methods

Methods are highly diverse: participatory design and workshops (Youth as Peer Auditors, WeAudit, Sociotechnical AI Governance workshop), qualitative case studies (A Sociotechnical Audit of facial recognition), framework development grounded in STS theory (iSTS Framework, Looking under the Bonnet), decolonial and critical theory (Decolonial AI), and mixed-methods combining technical evaluation with stakeholder analysis (Sociotechnical Safety Evaluation). There is a strong emphasis on stakeholder engagement and participatory approaches, which distinguishes this category from the technical and organizational buckets.

What do they miss?

Despite the breadth of ambition, many sociotechnical papers remain at the level of frameworks, principles, and recommendations without providing concrete, replicable audit methodologies. The tension between comprehensiveness and operationalizability is acute: papers that try to account for everything (power, context, stakeholders, history, environment) often struggle to specify what an auditor should actually do on Monday morning. Additionally, the category includes several papers that are really about sociotechnical AI governance broadly rather than auditing specifically, which dilutes the focus. There is also a risk of the sociotechnical label becoming a catch-all that obscures meaningful distinctions between very different kinds of work.

2.5 Industry Frameworks (4 papers)

What do these papers cover?

Four papers represent industry and professional body perspectives: PwC's Responsible AI framework, KPMG's Trusted AI Approach, the IIA's AI Auditing Framework, and the proposal for AI Audit Standards Boards. The Big 4 papers (PwC, KPMG) provide practitioner-oriented frameworks for operationalizing responsible AI within organizations, emphasizing principles, governance pillars, and maturity models. The IIA framework targets internal auditors specifically, helping them understand AI risks and identify controls. The Standards Boards paper argues that current approaches (internal auditing, industry-led standards, regulation) are all insufficient and proposes a dedicated standards-setting body modeled on financial accounting.

What do they miss?

Industry frameworks tend to be high-level and principle-based, offering maturity models and governance structures without specifying technical evaluation methods. They are oriented toward organizational risk management rather than identifying or measuring specific harms. There is limited engagement with academic research on fairness, bias, or sociotechnical systems. The PwC and KPMG papers, while practically useful, also carry inherent conflicts of interest: as consulting firms that both design and audit AI systems, their frameworks may prioritize client relationships over independent scrutiny.

2.6 Other (13 papers)

The 'Other' category collects papers that provide important conceptual or ethical foundations but do not focus specifically on auditing. These include: influential ethical frameworks (AI4People, A Unified Framework of Five Principles, Principles alone cannot guarantee ethical AI), evaluations of ethics guidelines (The Ethics of AI Ethics), scoping reviews of GenAI ethics (Mapping the Ethics of Generative AI), and papers on adjacent topics like meaningful work (The Ethical Implications of AI For Meaningful Work), decentralized AI governance (Trustless Autonomy), and AI organizational integration (Rising with the machines, Managing Artificial Intelligence). While not audit-specific, many of these papers provide the normative foundations that audit frameworks draw upon.

3. Cross-Category Patterns and Gaps

3.1 Where is the literature over-concentrated?

The most striking concentration is in sociotechnical framing papers. Fully a third of the collection articulates why auditing should be sociotechnical, what a sociotechnical perspective entails, or proposes conceptual frameworks for sociotechnical governance. This reflects the current moment in the field: there is strong consensus that purely technical audits are insufficient, and substantial intellectual energy has gone into articulating alternatives. However, this consensus has outpaced the development of concrete, empirically validated sociotechnical audit methodologies.

There is also significant concentration in framework proposals and literature reviews. Many papers propose audit frameworks, taxonomies, or typologies without testing them empirically. The field appears to be in a framework-proliferation phase, with many competing proposals and limited convergence on shared standards or methods.

3.2 Where is the literature thin or missing?

Post-deployment auditing receives limited attention. Most papers focus on pre-deployment evaluation or propose frameworks for the development lifecycle. How audits should work for systems already in production, how to detect emergent harms over time, and how to handle model drift and changing user populations are underexplored.

Supply chain and ecosystem auditing is nearly absent. Only ‘The case for a broader approach to AI assurance’ explicitly addresses upstream harms in the AI supply chain (environmental impact, labor exploitation, data sourcing). As AI systems increasingly rely on third-party models, datasets, and infrastructure, the question of how to audit across organizational boundaries becomes critical.

Empirical validation of audit methods is thin. The field has many proposed frameworks but few rigorous evaluations of whether they work. Notable exceptions include the AstraZeneca case study (Operationalising AI governance), the WeAudit user study, and the Youth as Peer Auditors workshop. Most framework papers remain purely theoretical.

Global South and non-Western perspectives are largely absent. The regulatory papers are EU-centric, the industry papers are from Western consultancies, and the academic papers overwhelmingly come from North American and European institutions. Decolonial AI is a notable exception that explicitly addresses this gap.

3.3 Implicit definitions of ‘good auditing’

Across the categories, several implicit standards emerge for what constitutes good auditing, though these are rarely made explicit:

- **Comprehensiveness:** Good audits should cover technical, organizational, and societal dimensions (the sociotechnical consensus).
- **Independence:** Good audits should involve external or independent parties, not just internal teams (Broken Bus, Standards Boards, regulatory gap papers).

- **Transparency:** Good audits should produce transparent, interpretable reports (Audit Cards, From Transparency to Accountability).
- **Stakeholder inclusion:** Good audits should incorporate perspectives of affected communities (WeAudit, Youth as Peer Auditors, Decolonial AI).
- **Actionability:** Good audits should lead to concrete changes, not just reports (Broken Bus, Towards AI Accountability Infrastructure).
- **Continuity:** Good audits should be ongoing rather than one-time events (Sociotechnical Safety Evaluation, multiple lifecycle papers).

However, these criteria are in tension with each other. Comprehensive, stakeholder-inclusive, continuous auditing is enormously resource-intensive, and the literature provides limited guidance on how to prioritize or make tradeoffs.

3.4 Academic vs. industry divergence

Scope of concern: Academic papers increasingly frame AI auditing as a matter of social justice, democratic governance, and power redistribution. Industry frameworks focus on organizational risk management, compliance, and reputation protection. The Big 4 papers do not engage with concepts like decoloniality, epistemic justice, or community-led auditing.

Level of abstraction: Academic papers tend toward conceptual frameworks and theoretical arguments. Industry papers provide more concrete (if shallower) guidance: maturity models, checklists, governance structures. This creates a complementarity problem where neither literature alone is sufficient.

Accountability mechanisms: Academic papers emphasize external accountability (to publics, regulators, affected communities). Industry papers emphasize internal accountability (to boards, risk committees, compliance teams). The gap between these two visions of accountability is a central tension in the field.

3.5 Key tensions

Technical rigor vs. real-world relevance

The technical auditing papers offer the most methodologically rigorous approaches (statistical fairness metrics, hypothesis testing frameworks, scaling laws), but they evaluate models in idealized conditions that may not reflect deployment realities. Conversely, sociotechnical papers capture real-world complexity but often lack the methodological precision needed to produce replicable, comparable results. The field has not yet found a satisfying integration of quantitative rigor and contextual sensitivity.

Standardization vs. context sensitivity

There is a fundamental tension between the desire for standardized audit procedures (which enable comparability, regulatory compliance, and professional credentialing) and the recognition that AI harms are deeply contextual (what counts as fair or harmful depends on the specific application, population, and institutional setting). The regulatory papers push toward standardization; the sociotechnical papers push toward context sensitivity. The Audit Cards

paper represents one attempt to bridge this gap by standardizing reporting while allowing contextual variation in methodology.

Pre-deployment vs. post-deployment

The overwhelming focus is on pre-deployment evaluation: assessing models before they are released. Post-deployment monitoring, ongoing auditing of live systems, and retrospective assessment of realized harms receive far less attention. This is particularly problematic for generative AI systems, where behaviors emerge from complex interactions with users and other systems that cannot be fully anticipated before deployment.

Access and power

Multiple papers identify access as a fundamental constraint on auditing. Black-Box Access is Insufficient and Access Denied demonstrate that limited access undermines audit quality. From Transparency to Accountability proposes frameworks for determining appropriate access levels. But the question of who has the power to grant or deny access, and what legal or institutional mechanisms could compel meaningful access, remains largely unresolved. The regulatory papers gesture toward solutions (the DSA and AIA mandate certain access provisions), but implementation details are sparse.

4. Summary and Implications for the Working Group

This literature paints a picture of a field in transition. The intellectual case for sociotechnical AI auditing is well established. There is broad consensus that auditing must go beyond technical model evaluation to encompass organizational practices, stakeholder impacts, and societal consequences. However, the field has not yet translated this consensus into mature, empirically validated, practically implementable audit methodologies.

Several priorities emerge for the working group:

- **Develop concrete audit protocols.** The field needs fewer frameworks and more detailed, tested protocols that specify what to do, how to do it, and how to report results. The Audit Cards proposal and the three-layered LLM auditing approach are steps in the right direction.
- **Bridge the academic-industry divide.** Neither the academic nor industry literatures are self-sufficient. The working group is well-positioned to develop approaches that combine academic rigor with practical implementability.
- **Tackle post-deployment auditing.** The literature is heavily pre-deployment focused. Methods for ongoing monitoring, harm detection, and retrospective assessment of deployed systems are urgently needed.
- **Confront the access problem.** Effective auditing requires meaningful access to models, data, and organizational processes. The working group should engage with the legal and institutional mechanisms that could make such access possible.
- **Include missing voices.** The literature is dominated by North American and European perspectives. Engaging with Global South scholars and affected communities would strengthen both the legitimacy and the substance of audit approaches.